



HYPERTROPHY: **VOLUME**



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HYPERTROPHY: VOLUME

MINI-EBOOK

One of the most mystifying and elusive questions in bodybuilding over the decades has been the following:

How many sets should you be doing to maximize hypertrophy?

Should you be doing ultra-high-volume training in Ronnie Coleman fashion? Or should you adopt a lower-volume approach (ala Dorian Yates)?

Should you keep your set volume stagnant through a training cycle? Or should you allow your volume to accumulate by adding additional sets throughout your mesocycle as recovery allows?

And what about soreness? Is it mandatory for an effective workout, or is it simply one of many tools we can use to help guide us in our pursuit of pure mass?

If you've ever wondered similar things in the context of your own muscle-building endeavors, you've come to the right place.

In this mini-guide, we are going to explore the theory behind optimal set volume for maximum hypertrophy, the proxies for practical application, other important considerations and tips, as well as how to problem-solve any issues that may arise as you attempt to implement these principles.

Part I: The Theory of the Volume Landmarks

In hypertrophy training theory, there is a certain number of sets that will be too few to let you grow much or any muscle. You could consider this an insufficient amount of training volume for growth, and this is theoretically represented by a real number.

For example, imagine a seasoned bodybuilder doing one working set of bicep curls and leaving the gym not to return for five days.

If you were to guess, how much muscle growth do you think this lifter would experience? Without hesitation, most of us would likely say none.

But why?

The answer is simple: He's simply too advanced to be growing from just one set of barbell curls across the span of five days.



And, although most of us know this to be true due to basic observations of bodybuilding training protocols across the decades, many people are unaware of the terms that represent these physiological realities.

In other words, we know that most advanced bodybuilders can't grow any detectable amount of muscle by doing one set of bicep curls across the span of five days, but why is that?

And how can we speak about such ideas in a systematic way that corresponds to evidence-based concepts that may be difficult to quantify in real time and space?

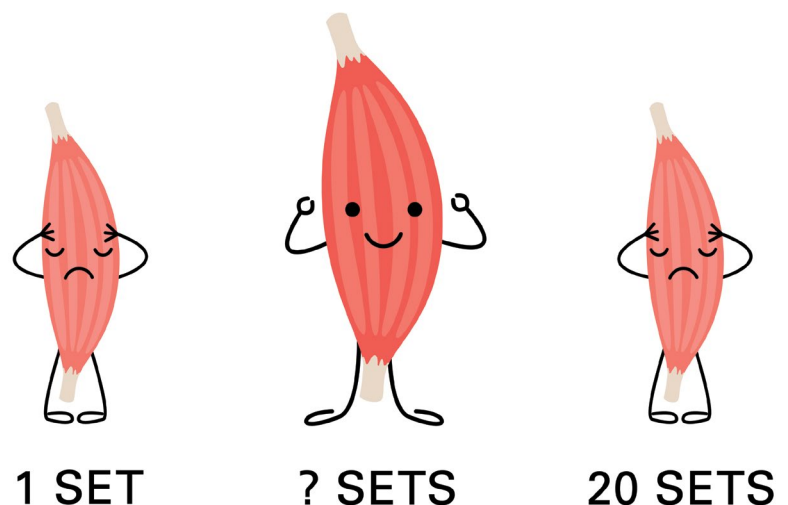
In an attempt to create new language around how to train with an appropriate number of sets for maximum hypertrophy, we've outlined some basic parameters known as the **volume landmarks**.

Returning to our example, in technical terms, we could say that his one set of barbell bicep curls is below his **Minimum Effective Volume (MEV)**, which is defined as the lowest number of sets per unit of time that leads to any net growth.

On the other hand, imagine a situation in which the same seasoned bodybuilder came into the gym and performed 20 working sets of barbell bicep curls in a single session.

In this case, we might be concerned that he would be *overwhelming* his muscles with so much work that they might not be able to recover properly by the next session, which could actually lead to a regression in size.

This top-end theoretical volume landmark has been termed the **Maximum Recoverable Volume (MRV)** and serves as a benchmark for the greatest number of sets per unit of time without exceeding acute recovery needs or violating long-term progression.



What does this mean?

The primary implication here is that you can have both *too much* of a good thing and also *not enough* of a good thing to make your efforts worthwhile.

The Gold Bars Analogy

If we were to load your car with a tenth of a gram of gold, it might not even be worth it for you to drive to the bank and deposit it since the fuel alone would cost you more than the task itself is worth.

In training terms, this would be considered a sub-MEV move.

But, if we were to load 200 pounds of gold bars into your car, you would be rich once you drove to the bank and deposited the gold, and that's great!

In training terms, this would land you in a very effective and productive range of set volume between your MEV and your MRV.

However, if we were to load 2,000 pounds of gold bars into your car, your suspension might collapse and you might not be able to drive the gold to the bank at all, making the entire endeavor a net negative since you'd now be left to repair your vehicle without any of the monetary gains from the gold.



In much the same way, performing additional sets beyond your MRV could cause more damage than growth.

All in all, the point here is that it's very important to know how many sets you should be doing for each muscle group in a given workout such that the sets fall strategically

between your MEV and MRV.

When done consistently, this will allow you to grow muscle as effectively as possible over the long-term.

Part II: The Proxies

How do we know if we're under, at, or over our MEV or MRV values?

Although we can't always tell precisely, our bodies can give us clues if we pay proper attention.

We know that workouts that give our target muscles a massive pump, make our muscles temporarily very weak and unstable, and make our muscles sore and tired in the hours and days after likely also stimulate muscle growth.

And there does seem to be a relationship between your session volume and the extent to which you experience these muscle-building proxies.

On a fundamental level, the more sets you do in a given session for a given muscle group, the more "stimulus proxies" you tend to experience. So in some sense, more *really* is better as long as the targeted muscles can recover properly before they are scheduled to be trained again.

If you feel strong and recovered by the time you have to train again, you'll be able to stimulate robust growth again.

On the other hand, if you do too many sets to the point that you can no longer recover on time, you likely won't grow as much muscle. And, as we alluded to before, it's possible that you could actually hinder your muscle-building efforts by performing a number of sets well beyond your MRV.

Fortunately, we can detect and use the following proxies of stimulus and recovery to help guide us toward an optimal number of sets to promote maximum muscle growth each time we train.

Pump and Perturbation

The first proxy to consider is what many of us in the bodybuilding community consider “the pump”

For those who may be new to the term, the pump is simply a notable swelling and tightening of the target muscles.



There is good reason to believe that, in most cases, you should feel *some* degree of a pump in the target muscle to surpass your MEV, assuming your goal is to grow as much muscle as possible.

In other words, if you fail to perceive any degree of a pump at all, it’s probably a good idea to perform another set and reassess the magnitude of the pump.

On the other hand, if you have an undeniably gnarly, skin-splitting pump, it's typically a good idea to stop training that muscle for that session, as doing much more could cause more damage than growth.

Can you still heal from excess damage incurred by an unnecessarily aggressive training stimulus? Absolutely. But you likely won't grow as much muscle as you would compared to a situation in which you did fewer sets and stopped once the pump was profound.

And this is possibly one of the most misunderstood concepts in bodybuilding. Why? Because, at this point, more is no longer better because you have likely exceeded a productive level of volume for that particular muscle group in a single session. And this is a very important consideration when it comes to developing the skill of autoregulating single session volume for best long-term results.

Secondly, the term perturbation refers to a breakdown in the target muscle's normal sensation or operation abilities.

Feelings of weakness, cramping, poor proprioception (difficulty coordinating normal movements like walking or grabbing), and sporadic muscle failure are all indicators that the target muscle has been significantly perturbed. In these cases, such perturbation is indicative of a very powerful growth stimulus having been delivered.



Perturbation can be even more powerfully confirmed if you're experiencing these strange sensations while also experiencing a huge dropoff in acute strength during that time.

In other words, if your biceps feel strange and relatively light weights feel awkwardly heavier than they should, you can consider your biceps properly perturbed! However, if you don't get *any* of these signs during a workout, you might consider pushing forward with more sets for that given muscle if the other proxies appear to be permissive as well.

If, however, your biceps are so wobbly that you have trouble putting the weights back, it's almost certainly time to discontinue training your biceps for that session.

In conclusion, you can use the proxies of pump and perturbation to help you determine how many sets to perform during your workout.

In addition to determining an optimal number of sets in this particular way, we have two other proxies that are used after the workout has already been completed.

Disruption and Strength Perception

Disruption is the term we use to refer to muscle damage of any kind after a workout.

We can say that a muscle was significantly disrupted if it feels anywhere from a bit tighter than usual in the hours after the workout all the way up to a debilitating level of soreness that doesn't reach peak tenderness until a day after your workout. In the bodybuilding realm, this highest level of post-workout soreness is referred to as **delayed onset muscle soreness** (DOMS).

If you complete an entire workout for a muscle and you can sincerely report that you feel *nothing* in the hours and days after completion, then you can be relatively sure that the disruption you experienced was minimal and sup-optimally effective.

At that point, if all other variables are equal, you can (and should) consider adding more sets to that same workout the next time you are scheduled to perform.

Let's use a quick example to illustrate this point.

Let's say it's the first week of a new training cycle, and you've programmed two sets of hack squats and two sets of leg extensions for your Monday leg session. If Tuesday and Wednesday roll on by like a tumbleweed in the wind without a hint of muscle soreness, you can be confident that you can add sets to your next Monday session. In this case, it might be appropriate to add a single set of hack squats to your regimen while keeping your leg extensions the same. Now, if Tuesday and Wednesday arrive during your second week of training bearing gifts of noticeable muscle soreness, you can be reassured that you made a good decision to increase your leg volume in your Monday session.

On the other hand, if you're still sore from a workout by the time it's appropriate to train that same muscle again a few days later, it's likely that you experienced a level of disruption that could interfere with growth. Using the above example, if your second leg day of the week is on Thursday, and you're still sore on Thursday from when you trained legs on Monday, you probably overdid it on Monday.

In this situation, you would be wise to maintain or *reduce* how many sets you do in that workout next week such that you are not sore at all when you train that muscle group again, but definitely don't increase sets. This is the sweet spot when it comes to programming from session to session. The goal is to be recovered each time you enter the gym to train any given muscle group.

This, however, begs another question: Is soreness mandatory for growth?

Soreness is not mandatory, but your muscles still need to heal on time for the next workout. If they don't heal on time, you've probably done too many sets.

But how can you know if you're recovered?

In the moments before a muscle-growth training session, you should feel strong and able to train the target muscles hard.

If the target muscles still feel weak and under-recovered, you probably exceeded an appropriate level of volume during the session in which you last trained them and could benefit by doing fewer sets next week in that same workout, or at least not adding any, and allowing your natural adaptation to training volume week over week to catch up and

let you heal on time after next week's workout.

Returning to our hack squat and leg extension example, this means that, if four sets of hack squats and four sets of leg extensions left you crippling sore by Thursday when you were supposed to do leg presses and barbell reverse lunges, you would be wise to be more conservative the next time you do hack squats and leg extensions.

On the other hand, if you feel *ultra* strong and can't even recall feeling weak after your last workout, your recovery is clearly abundant and you can consider adding a set to that workout next week.

In an ideal situation, you should be noticeably fatigued from the last workout through the perception of significant perturbation and maybe even disruption, yet heal just on time for the next workout so that you can no longer be sore, feel strong, perform well, and stimulate growth once again.

Part III: Practical Applications

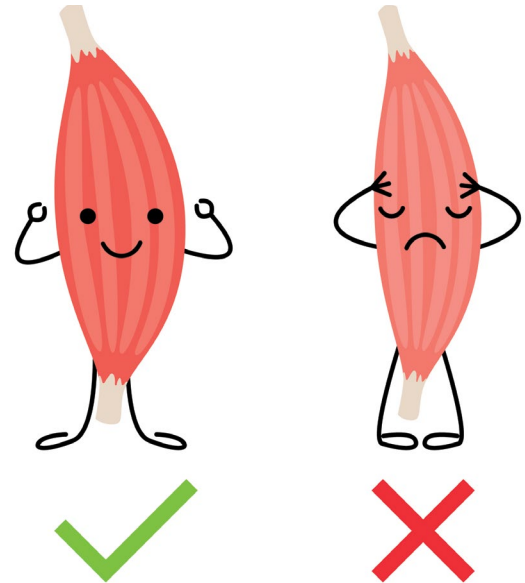
If you want to get the most out of your training, it's a good idea to use all of the stimulus and recovery proxies to regulate the number of sets you do.

How many sets you add or subtract will be based on the information you derive from our four aforementioned proxies.

From here, let's take a look at how these informational proxies can tell you what to do about your training volume.

For the sake of simplicity, we've condensed the main ideas into five concise points.

1. If you don't have a robust pump or any notable perturbation within a session, perform



more sets for that muscle group until you do.

2. Once you have a good or even great pump or (notable or even debilitating perturbation), resist the temptation to do additional sets for that muscle group in that session.
3. If you hardly get disrupted at all, and you feel ultra-recovered from the workout by the time your next session for that muscle comes up, feel free to add a set to the plan next week.
4. If you get notably disrupted and manage to recover on time to have another good workout, your best bet is to keep the volume stable for the time being.
5. If you don't heal from soreness or feel much weaker than normal by your next session, consider reducing the volume of next week's corresponding workout by a set or two in order to make sure your recovery abilities aren't exceeded..

It really can be just about this simple.

In fact, if we were to summarize the entirety of volume autoregulation from session to session, we might say this:

Do as many sets as it takes to get a good pump and disruption, stop, and then evaluate how sore and weak those sets make you in the following hours and days. From there, make any necessary adjustments based on the feedback you receive from perceived soreness and weakness in your next workout later that week for that same muscle group, execute that new level of single session volume the following week, and repeat the process over and over again across your mesocycle.

But what about managing session volume across multiple exercises?

Up until now, we've been talking about adding sets and subtracting sets, but what if we do multiple exercises for a single muscle group in a session? After all, we mentioned the combination of hack squats and leg extensions earlier, and most people tend to train their muscle groups across multiple exercises rather than a single movement.

In general, we can most benefit from 1-3 exercises per muscle per session, and usually no more than 2.

However, introducing multiple exercises per muscle group introduces several new questions for consideration when it comes to nuancing volume.

For example, which exercises should we add sets to or subtract sets from?

Simply put, the answer is whichever ones you feel like.

In reality, it's a bit more nuanced than this, but not by much. What you want is to do more sets of movements that *really* hit the target muscle and relay sensations of tension, burn, pump, and perturbation while being easy on the joints and minimally draining in a psychological sense.

Similarly, if you have to subtract sets away, you'll want to do this from exercises that don't seem to provide as much of a stimulus, beat up your joints a bit more, or are perceptually harder on your mind (the exercises that are the most psychologically intimidating to perform).

So if you have to add or subtract sets next week, you'll want to ask yourself the question:

"Which exercises should I add/subtract to/from in order to get the best combination of high stimulus with the lowest amount of fatigue?"

For example, if you did three sets of leg presses and two sets of squats last Monday, and, via the proxies on Monday and Thursday (your second quad workout of the week), you decided to add a set, ask yourself if adding a set of leg presses or squats would give you the best combination of stimulus to fatigue in that workout.

If you think four sets of leg presses will make the workout stale, add the spare set to squats. If you think three sets of squats will beat up your lower back more than four sets of leg presses would for the same stimulus, then add that set to leg presses instead. As the saying goes, it's not rocket science.

Part IV: Additional Considerations

Be conservative with your initial session volumes.

Because fatigue accumulates and takes time to reduce, doing too much volume early in

your training cycle gives you the downside of suboptimal growth with the baggage of unwanted fatigue you will have to deal with later.

On the other hand, doing too little volume still gives you a bit of growth but without the baggage of extra fatigue.

For this reason, starting on the low end of volume is usually a wise idea. In most cases, beginning with 2-6 working sets per muscle group per session spread over 1-2 exercises is best practice.

If you've been training for a while and have a better idea of how many sets will give you a good pump and perturbation for a given muscle, by all means, implement your knowledge!

But, if you don't have much or any experience to base your volume estimates on, starting at 2-6 sets per muscle per session is highly recommended.

Take proper notes to quantify stimulus proxies and alter your volume systematically.

Once you do a session, note all of the stimulus and recovery proxies during and after it, and then alter your volumes session by session and week by week. Although this may seem like additional work, it's certainly the most systematic way of going about making educated tweaks to your programming as you pursue an increasingly muscular physique.

If you follow the process described in the fifth point above, you'll be well on your way to your best volumes and best gains.



The only big tips are to be honest with yourself and err on the side of less if you have to make what seems to be an unclear choice.

It's almost always better to do fewer sets with greater quality than to do more sets with lesser quality.

Be aware of hitting or eclipsing your MRV.

Once you've been training for a few weeks, you'll have gotten the hang of adding and subtracting set numbers to align your growth stimulus needs with your recovery abilities.

During this time, however, fatigue from the entirety of your training program will be accumulating. At some point, it will rise so high that you won't be able to progress in rep strength anymore, and you may actually backslide in strength.

Once that occurs, you're likely close to or at your MRV, and it's time to back off and let your body recover before making another hard push.

If you do two sessions in a row for the same muscle group and fail to improve over last week's performance in those sessions, it's time to back off and let fatigue drop, which leads us to our next point.

Remember to deload and cycle your volume appropriately.

After you're no longer able to increase your performance, you'll need about a week of much easier training to recover.

Known as a **deload week**, this week is best programmed with about half of the reps, half of the sets, and even half of the weight used in your last hard week of training.

Sounds super easy, right? It is!

But it's during this ultra-easy week of training that your body and muscles recover to a huge extent, readying you for another month or two of hard training to come.

Once you're done with the deload week, you can rebuild your plan with a few different

movements and sequences of muscles trained, and go back down to 2-6 working sets for each muscle while repeating the process of volume adjustment while training hard and reaping even more muscle gains.

This should work incredibly well when done properly, but if you do happen to run into problems adjusting your volumes, we might be able to help.

Here are a few common problems with volume autoregulation and how to address them productively.

Part V: How to Address Common Problems

Problem 1: I don't get pumps.

Some people report that they either don't get notable pumps or can't reliably tell when they have a good pump or not.

If you're one of these people, don't worry.

Your best bet is to try to determine how much perturbation the muscle has experienced and then regulate your volume based primarily on that specific feedback modality.

For example, if your triceps feel totally fine after a few sets, keep training them. If they start to feel weak and wobbly, *stop* training them and adjust your tricep volume next week in order to preserve your recovery abilities for your next tricep workout.

Problem 2: I can't tell if I'm detecting perturbation

Some folks have trouble telling if their muscles have been perturbed, and that's fine.

If you're in this group, you can simply go by pumps alone and use them to alert you when enough is enough in a given session.

If you can't tell either how your pump or perturbation is, just stay on the low end of the

2-6 set recommendation, and wait to see how you recover for a session for that muscle later in the week.

If you're *super* recovered, add some sets in your next session. If you're *barely* recovered, keep the sets the same. And, if your muscles still feel *mega-trashed*, obviously reduce the number of sets in the following workout.

Problem 3: I don't get sore.

Some people just don't experience soreness in certain muscle groups, and that's not a big deal either.

If you don't get sore, your best bet on regulating volume is to go based on strength recovery. If you feel strong before your next session for that muscle, you can be relatively certain the volume you did in your last session wasn't overkill. If you feel unproductively weak, consider reducing the volume. If you're not sure how you feel, just try doing that next workout, and let your performance results dictate your responses. If you hit PRs easy, you're very recovered. If you are so weak you actually underperform compared to last week, it's time to reduce that volume a bit.

Just remember, high-quality training will bang you up and make you feel weak in the moments after your training but lets you heal on time to allow you to feel strong and recovered before your next training session. If you achieve that, no soreness detection of any kind is required, which should be an encouraging bit of information for anyone who has ever been under the impression that soreness is mandatory in terms of quantifying an effective workout for muscle growth.

Part VI: *Conclusion*

At the end of the day, we can extract some very simple advice from the specifics on volume regulation outlined in this guide.

But, what's most important to remember is this:

- If you're recovering easily and ahead of schedule, add a set to next week's workout.
- If you're getting nice and sore but recovering on time, don't change your volumes.
- And, if you're getting way too beat up to recover on time, subtract a set or two from your next week's plan, and go from there.

There are no universally applicable magical set numbers or shortcut solutions. The key is simply to pair your growth stimulus needs strategically to your recovery abilities in a way that optimizes training performance while managing fatigue on a week-to-week basis.

